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Oefeningenreeks 2D nr. 14.13

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{2(1 - \cos x)}{x^2} \\ &= \lim_{x \rightarrow 0} 4 \cdot \frac{\sin^2\left(\frac{x}{2}\right)}{4 \cdot \left(\frac{x}{2}\right)^2} \\ &= \lim_{x \rightarrow 0} \left(\frac{\sin\left(\frac{x}{2}\right)}{\left(\frac{x}{2}\right)} \right)^2 \\ &= 1 \end{aligned}$$

References